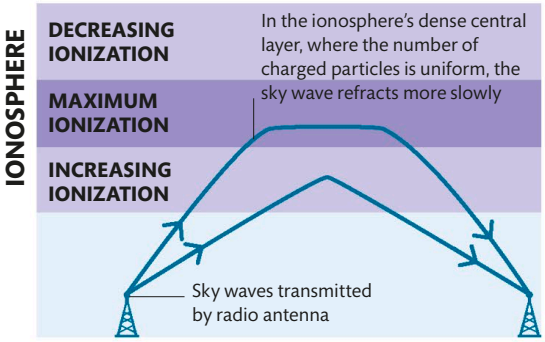
Radio signals

Radio waves are used to send and receive information across distances without wires or cables. We rely on radio signals for broadcasting, telecommunications, navigation, and computer networks.

Sending signals

Radio waves can contain information such as sound, text, images, and location data. This information is encoded into the wave by modifying different wave features, such as its frequency or amplitude (see opposite). To send information between locations, the radio signals are emitted by a transmitter using an antenna and travel through the air until they are picked up by receivers, also using antennas.



Refraction in the ionosphere
When a sky wave is transmitted into the ionosphere, the electrically charged layer of Earth's atmosphere, it bends (refracts). The degree to which it is refracted is influenced by the angle of the wave, the wave's frequency, and the number of charged particles present in the layers of the ionosphere.

